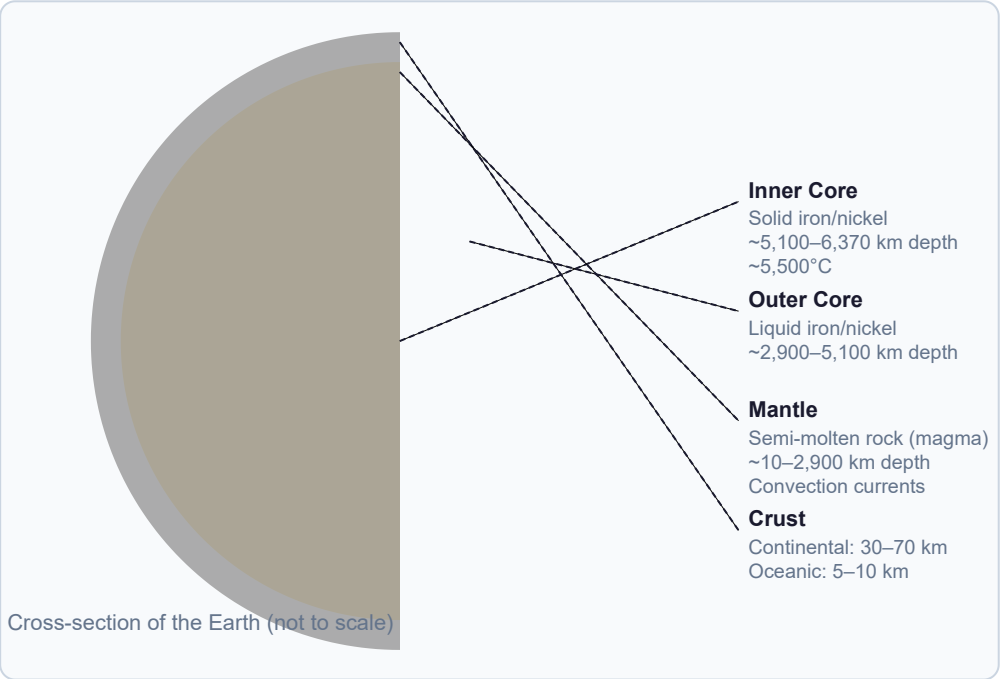
RAG Learning

Eduqas GCSE Geography (C/4) — Component 3

3.1 Tectonic Hazards

1. Structure of the Earth

The Earth is made up of four concentric layers, each with different properties:



Layer	Thickness	State	Composition	Temperature
Crust (Continental)	30–70 km	Solid	Granite (silica-rich)	~300–500°C at base
Crust (Oceanic)	5–10 km	Solid	Basalt (denser)	~200–400°C at base
Mantle	~2,890 km	Semi-molten (asthenosphere)	Peridotite (silicate rock)	1,000–3,700°C

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Outer Core	~2,200 km	Liquid	Iron and nickel	~4,000–5,000°C
Inner Core	~1,270 km radius	Solid (high pressure)	Iron and nickel	~5,100–5,500°C

Convection currents in the mantle are the driving force behind plate movement. Hot magma rises, spreads laterally, cools, then sinks again — dragging tectonic plates with it. This process is called **seafloor spreading** at mid-ocean ridges.

2. Plate Tectonics

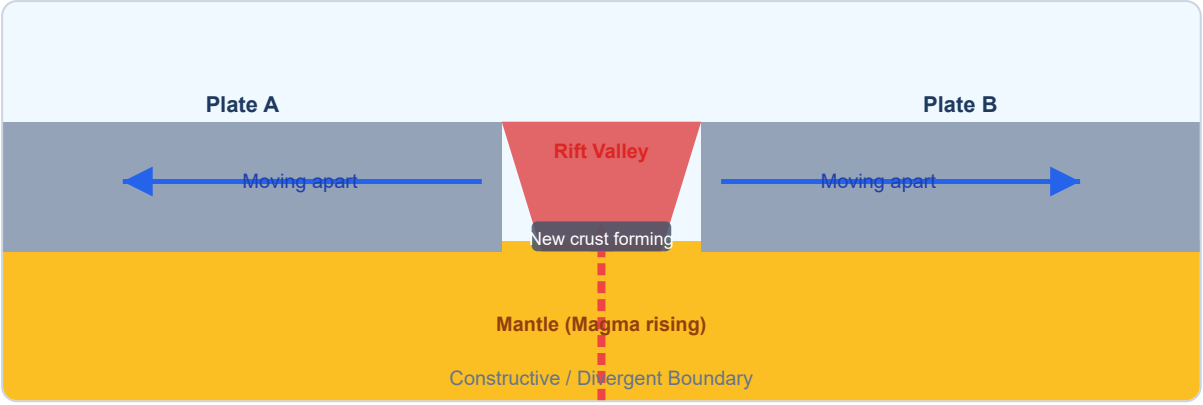
Theory of Continental Drift (Wegener, 1912)

Alfred Wegener proposed that today's continents were once joined as a supercontinent called **Pangaea**. Evidence:

- **Jigsaw fit** — coastlines of S. America and Africa fit together remarkably closely.
- **Fossil evidence** — identical fossils of Glossopteris (a fern) and Mesosaurus (a freshwater reptile) found on opposite sides of the Atlantic — impossible if they had always been separated by ocean.
- **Rock matching** — identical rock sequences found in Brazil and West Africa.
- **Mid-ocean ridges** — new oceanic crust forms at mid-ocean ridges as plates pull apart.
- **Paleomagnetism** — alternating bands of magnetic polarity on the seafloor either side of mid-ocean ridges prove seafloor is spreading symmetrically.

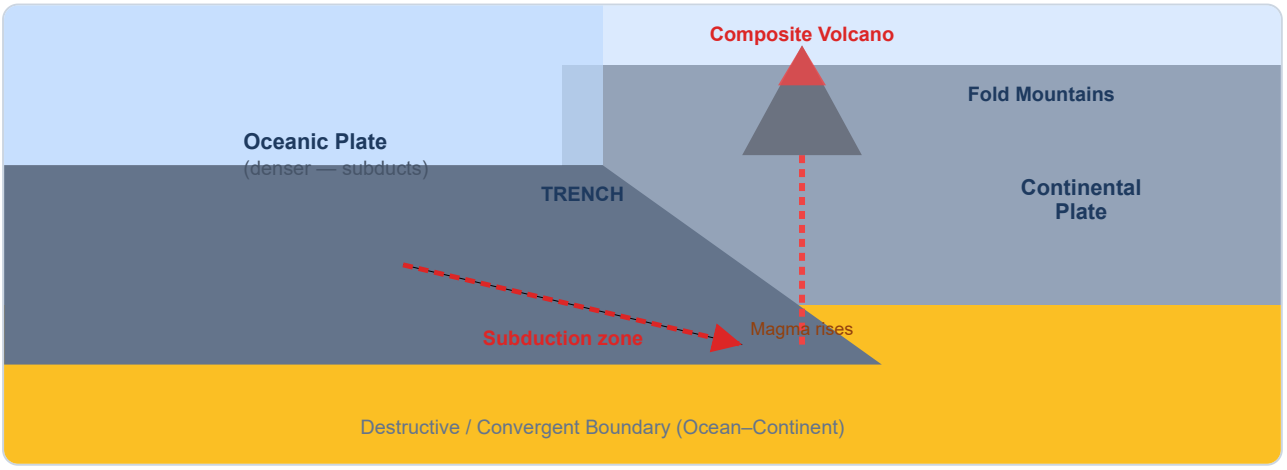
> **Tectonic plates** — rigid segments of Earth's crust and upper mantle (lithosphere) that float on the semi-molten asthenosphere. Major plates: Eurasian, North American, South American, African, Indo-Australian, Pacific (largest), Antarctic. Several smaller plates: Juan de Fuca, Caribbean, Philippine, etc.

3. Constructive (Divergent) Plate Boundaries



- Plates move apart; magma rises through the gap and solidifies, forming **new oceanic crust**.
- Process creates: **mid-ocean ridges** (underwater mountain chains), **rift valleys**, and **shield volcanoes**.
- **Mid-Atlantic Ridge** — Iceland sits on this ridge. Iceland experiences frequent volcanic eruptions (Eyjafjallajokull 2010; Fagradalsfjall eruptions 2021–2023) due to its position on the constructive boundary.
- **East African Rift Valley** — continental divergence creating a rift valley (Lakes Tanganyika, Malawi, Turkana).
- **Volcanoes:** Shield volcanoes — wide, gently sloping, basaltic (low-viscosity) lava, less explosive. Example: Mauna Loa, Hawaii; Etna (partly).
- **Earthquakes:** Generally shallow focus and low magnitude — less destructive.

4. Destructive (Convergent) Plate Boundaries



Ocean-Continent (Subduction)

- Denser oceanic crust subducts beneath lighter continental crust into the mantle — forms a **subduction zone** and **deep-sea trench**.
- Oceanic crust melts in the Benioff zone; magma is forced upward through the continental crust, forming **composite (stratovolcanoes)**: steep, explosive, andesitic lava (high viscosity). Examples: Andes (Chile, Peru), Cascade Range (Mt St Helens, USA — erupted 1980), Mt Pinatubo (Philippines — erupted 1991).
- Earthquakes: deep focus (Benioff zone), high magnitude — very destructive.

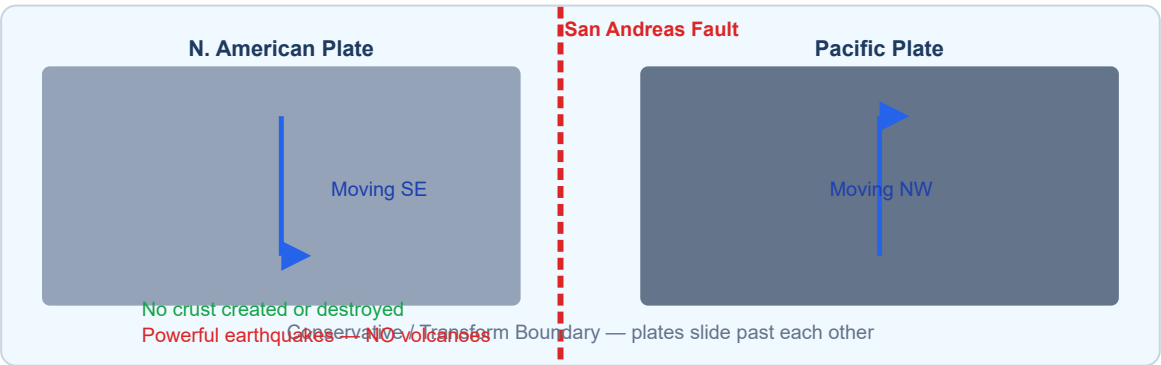
Ocean-Ocean Convergence

- One oceanic plate subducts under the other. Forms: deep-sea trenches (Mariana Trench — deepest point on Earth, 11,034m) and **island arcs** — chains of volcanic islands (Japan, Caribbean islands).

Continent-Continent Convergence

- Neither plate subducts (both are too buoyant). Instead, crust crumples and folds upward to form **fold mountain ranges**. No volcanoes. Powerful earthquakes.
- Example: **Himalayas** — Indo-Australian plate colliding with Eurasian plate. Still rising ~5mm/year. Nepal Earthquake 2015 a result of this collision zone.
- Alps — African plate colliding with Eurasian.

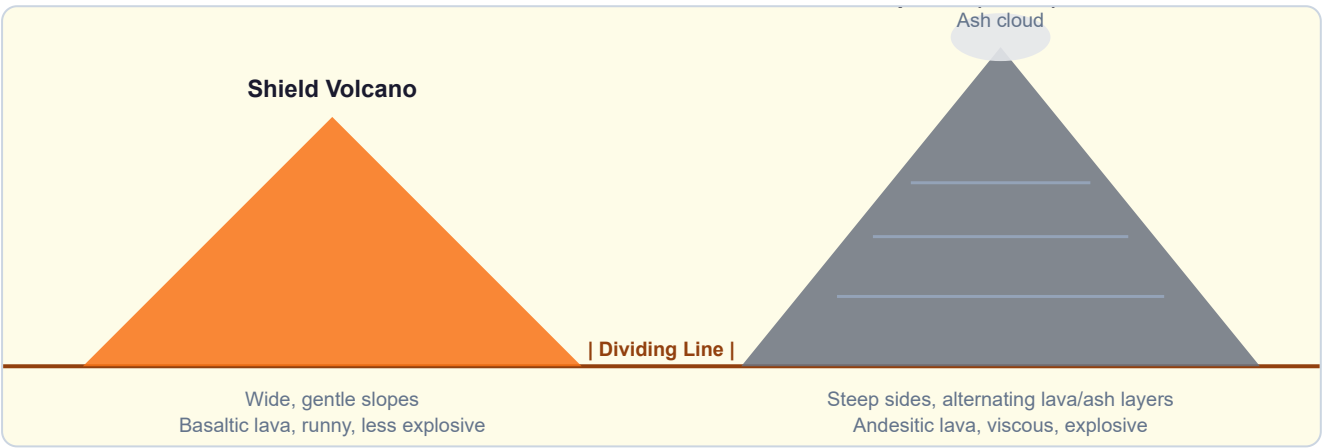
5. Conservative (Transform) Plate Boundaries



- Plates slide past each other horizontally — no crust created or destroyed.
- **No volcanic activity** (no subduction, no magma generation).
- Friction builds between plates — when released, produces **very powerful earthquakes** (shallow focus, extremely destructive near populated areas).

- **San Andreas Fault, California** — North American plate vs Pacific plate (Pacific moving NW at ~5cm/year). San Francisco earthquake 1906 (estimated 7.9 magnitude). Los Angeles earthquake 1994 — Northridge, 6.7 magnitude, 57 deaths, $20bn damage.
- The Dead Sea Transform fault (Jordan/Israel/Palestine) is another example.

6. Volcanoes



Feature	Shield Volcano	Composite / Stratovolcano
Shape	Wide, gently sloping (1–10°)	Steep, cone-shaped (30–40°)
Lava type	Basaltic — low viscosity, runny	Andesitic/rhyolitic — high viscosity, thick
Eruption style	Frequent, less explosive (effusive)	Infrequent but explosive (pyroclastic)
Hazards	Lava flows (slow-moving)	Pyroclastic flows, ash fall, lahars, bombs
Boundary type	Constructive / Hot spot	Destructive (subduction zones)
Examples	Mauna Loa (Hawaii), Kilauea, Piton de la Fournaise (Réunion)	Mt St Helens (USA), Pinatubo (Philippines), Merapi (Indonesia), Etna (Italy)

Volcanic Hazards

- **Lava flows** — slow-moving (most people can escape), but destroy property and farmland.

- **Pyroclastic flows** — fast-moving (up to 700km/h), superheated (400–700°C) mixtures of gas, ash, and rock. Most deadly volcanic hazard. Killed ~29,000 in Mt Pelée, Martinique (1902).
- **Ash fall** — suffocates crops, collapses buildings, disrupts air travel (Eyjafjallajokull 2010 — 100,000 flights cancelled).
- **Lahars** — volcanic mudflows. Melted snow/ice mixes with ash. Armero, Colombia (1985): lahar from Nevado del Ruiz killed 23,000.
- **Volcanic gases** — SO_2, CO_2 toxic; can cause acid rain.
- **Tsunamis** — underwater eruptions can trigger tsunamis.

Benefits of Volcanoes

- **Fertile soils** — volcanic ash weathered to nutrient-rich soils. Java (Indonesia) is one of the world's most densely populated areas due to highly fertile volcanic soils.
- **Geothermal energy** — Iceland generates 25% of electricity from geothermal energy — heats 90% of homes.
- **Tourism** — Mt Etna, Kilauea, Yellowstone attract millions of visitors.
- **Minerals** — valuable minerals (copper, gold, sulphur) deposited around volcanoes.

7. Earthquakes

Focus — the point inside the Earth where an earthquake originates (where rock fractures/slips). Also called the hypocenter.
Epicentre — the point on the Earth's surface directly above the focus. Most severe shaking occurs here.
Seismic waves — energy waves that travel outward from the focus.

Types of Seismic Waves

- **P-waves (Primary)** — compression waves; travel through solid and liquid; fastest; arrive first on seismograph.
- **S-waves (Secondary)** — shear waves; only travel through solid; slower; cannot travel through Earth's liquid outer core.

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- **Surface waves (Love and Rayleigh)** — travel along Earth's surface; slowest but most destructive — cause most of the ground shaking and building damage.

Measuring Earthquakes

- **Richter Scale** — measures magnitude (energy released). Logarithmic: each point = 10x more energy. 0–2 (not felt) → 5–6 (moderate damage) → 7+ (severe) → 9+ (extreme, rare).
- **Moment Magnitude Scale (Mw)** — now preferred by scientists; similar logarithmic scale but more accurate for large earthquakes.
- **Mercalli Scale** — measures intensity (damage and shaking felt at a location). Rated I–XII. Subjective but useful where seismometers are absent.

Effects of Earthquakes

Primary Effects (immediate, caused directly by shaking)	Secondary Effects (caused by primary effects)
Buildings and bridges collapse	Tsunamis (if submarine earthquake)
Ground shaking and cracking	Landslides and avalanches
People killed/injured by collapsing structures	Liquefaction — waterlogged soil behaves like liquid, buildings sink (Christchurch NZ 2011)
Roads and railways buckle/break	Disease outbreaks (cholera, typhoid) from damaged sanitation
Fires from broken gas pipes	Homelessness → disease, exposure, mental health impacts

8. Tsunamis

A tsunami is a series of powerful ocean waves caused by a large underwater disturbance (earthquake, landslide, volcanic eruption). Most tsunamis are caused by **submarine earthquakes** — where the seafloor suddenly shifts vertically, displacing a vast column of water.

- **Deep water:** wave speed ~800 km/h, wavelength 100–200 km, amplitude just 1–2m (often unnoticed at sea).

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- **Shallow water (approaching coast):** speed slows dramatically but wave height increases enormously ("shoaling") — can reach 30–40m.
- Warning signs: sea withdraws rapidly from shore (draw-back) before the wave arrives.

2004 Indian Ocean Tsunami (Boxing Day Tsunami)

- **Cause:** 9.1 magnitude submarine earthquake (destructive boundary — Indo-Australian plate subducting under Burma plate) 250km off Sumatra, Indonesia.
- **Scale:** 230,000+ deaths across 14 countries (Indonesia, Sri Lanka, India, Thailand — worst). 1.7 million displaced.
- **Wave height:** Up to 30m in Sumatra; 6–9m in Sri Lanka/India.
- **Speed:** ~800 km/h across Indian Ocean — reached Sri Lanka in ~2 hours, Africa in ~7 hours.
- **Response:** Led to creation of Indian Ocean Tsunami Warning System (2006). Pacific Tsunami Warning Centre (PTWC) already existed since 1949.

9. Case Study 1: Nepal Earthquake, April 2015 (LIC)

Nepal Earthquake — 25 April 2015 | 7.8 Magnitude

Location: 80km NW of Kathmandu, shallow focus (15km depth), continent-continent destructive boundary (Indo-Australian plate colliding with Eurasian plate — Himalayas).

Primary Effects

- 9,000+ deaths; 23,000+ injured.
- 600,000+ homes destroyed (mostly mud-brick construction).
- 8,000+ schools and 1,000 health facilities damaged.
- Historic Durbar Square, Kathmandu (UNESCO World Heritage Site) — temples collapsed.
- Dharahara Tower (19th century) collapsed, killing 180 people sheltering nearby.

Secondary Effects

- Massive landslides — Langtang village buried (350 people), trekking routes blocked.
- Avalanches on Everest — 19 climbers killed at Base Camp; worst Everest disaster at the time.

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- 3 million people left homeless — displacement raised risk of disease (cholera, diarrhoea) as sanitation systems destroyed.
- Crops destroyed ahead of harvest — food insecurity.
- Tourism collapsed — 30% of GDP affected in the short term.

Why Was Impact So Severe? (LIC factors)

- **Poor building quality** — most rural housing built from unmortared stone or mud brick. No earthquake-resistant design. Urban buildings often poorly constructed despite regulations.
- **Remote mountainous terrain** — roads blocked by landslides; helicopter access only; aid reached many areas weeks late.
- **Population density** — Kathmandu Valley densely populated (1.5 million+), high exposure.
- **Poverty** — Nepal is one of Asia's poorest nations; GNI per capita ~$1,200. Limited government capacity to prepare or respond.

Responses

- **Immediate:** International aid poured in — India first responder (NDRF teams). USA, UK, China, Pakistan, EU — $4.4bn pledged internationally. UN OCHA coordinated. Nepali Army deployed search and rescue.
- **Short-term issues:** Aid bottleneck at Kathmandu airport (only one runway); looting of relief supplies; corruption allegations; political disputes delayed relief distribution.
- **Long-term:** "Build Back Better" reconstruction programme — earthquake-resistant design. By 2020 (5 years later), only ~65% of damaged homes rebuilt. Slow bureaucracy, land disputes, funding mismanagement criticised heavily by survivors and NGOs.

10. Case Study 2: Japan Earthquake and Tsunami, March 2011 (HIC)

Tohoku Earthquake and Tsunami — 11 March 2011 | 9.0 Magnitude

Location: 70km off Sendai coast, NE Japan. Destructive boundary — Pacific plate subducting under North American/Eurasian plate (Japan Trench). Strongest earthquake ever recorded in Japan; 4th

strongest globally since 1900.

Primary Effects

- 15,900+ deaths (mostly by tsunami, not earthquake — buildings withstood shaking well).
- 130,000+ buildings destroyed; 1 million+ damaged.
- Fukushima Daiichi nuclear power plant — tsunami overwhelmed sea defences (14m wave vs 10m sea wall), causing cooling systems to fail — nuclear meltdown in 3 reactors. Level 7 nuclear emergency (same as Chernobyl).
- Tsunami: heights up to 40m at some coastal points; travelled up to 10km inland; swept away entire coastal towns (Ishinomaki, Rikuzentakata).

Secondary Effects

- **Nuclear disaster:** 154,000 people evacuated from 20km exclusion zone around Fukushima. Long-term radiation contamination of soil and sea. As of 2023, exclusion zones still partially in place.
- Global nuclear energy debate reignited — Germany accelerated nuclear phase-out; Japan shut all 50 reactors for safety checks.
- Supply chain disruption — Japan produces 60% of world's silicon wafers for semiconductors; global electronics industry affected for months.
- Economic cost: ~$235bn — most expensive natural disaster in history.

Why Japan Coped Better (HIC factors)

- **Earthquake-resistant buildings** — strict building codes enforced since 1981 (and revised after 1995 Kobe quake): cross-bracing, shock absorbers, base isolators, automatic gas shut-off.
- **Early warning system** — Japan Meteorological Agency issued earthquake warning 90 seconds before strong shaking arrived; automated systems shut down trains and industrial machinery.
- **Tsunami warning** — coastal warning issued within 3 minutes; evacuation sirens activated.
- **Regular drills** — Disaster Prevention Day (1 September) — nationwide earthquake drills every year since 1960.
- **Organised emergency response** — Self-Defence Forces (100,000 deployed), pre-positioned emergency supplies, efficient hospital system.

BUT — Limitations Even for an HIC

- Tsunami overwhelmed sea walls designed for smaller events — underestimation of risk.
- Nuclear disaster was unexpected and unprecedented — containment plans inadequate.
- Evacuation traffic jams reduced escape rates in some coastal areas.
- Elderly population (Japan has world's oldest population) — struggled to evacuate quickly.

Comparison: Nepal (LIC) vs Japan (HIC)

Factor	Nepal 2015 (LIC)	Japan 2011 (HIC)
Magnitude	7.8	9.0 (much larger)
Deaths	9,000+	15,900+ (mainly from tsunami)
Building quality	Mud brick, poorly built	Earthquake-resistant to modern codes
Early warning	No effective system	90-second earthquake warning; tsunami alert in 3 min
Emergency response	Slow, disorganised, aid bottlenecks	Fast, organised, 100,000 military deployed
Long-term recovery	Still incomplete 5+ years later	Most areas rebuilt within 3–5 years
Main factor for deaths	Building collapse + LIC poverty	Tsunami inundation (not earthquake)

11. Managing Tectonic Hazards

Management falls into four broad categories: Prediction, Protection, Planning, and Aid.

Prediction

- **Volcanoes** — ground swelling (tiltmeters), increased seismic activity (seismometers), gas emissions (SO_2 and CO_2 levels rise before eruptions), thermal cameras detecting heat changes. Mt Pinatubo (1991) — successful 2-week warning: 60,000 people evacuated, saved thousands of lives.

- **Earthquakes** — cannot be predicted with sufficient accuracy for evacuation. Seismometers detect P-waves seconds before damaging S/surface waves arrive (seconds to ~1 minute warning). Long-term probabilistic hazard mapping identifies high-risk areas.

Protection (Engineering)

- **Earthquake-resistant buildings:** cross-bracing (steel frames), base isolators (rubber pads between building and foundations absorb shaking), counterweights (dampers on tall buildings — Taipei 101 has an 800-tonne pendulum), automatic gas shut-off, fire-resistant materials.
- **Tsunami sea walls:** Japan's Kamaishi sea wall (4km long, 63m deep foundations, 8m above sea level) — largest in the world at construction. The 2011 tsunami overtopped it but it reduced wave height and gave more evacuation time.
- **Breakwaters and coastal barriers** — Japan invested $12bn in tsunami sea walls along 400km of coastline after 2011.

Planning

- Land-use zoning — restricting development in high-risk zones near faults or low-lying coasts.
- Evacuation routes — clearly signed, regularly rehearsed.
- Regular drills — Japan (annual), California (Great ShakeOut — 10 million+ participants).
- Community education — knowing what to do in the first 72 hours (the critical "golden hour" window for survivors).

Aid

International response critical for LICs (Nepal, Haiti 2010). UNOCHA coordinates. Issues: logistics, corruption, dependency. Long-term reconstruction aid must focus on building back safer, not just rebuilding the same vulnerable structures.

> **Exam Tip:** For "evaluate management strategies" questions, use the framework: Prediction is *difficult for earthquakes* (but useful for volcanoes); Protection is *effective but expensive* (only HICs can fully implement it); Planning is *most cost-effective* and works in both HICs and LICs; Aid is *vital in the short term* but must avoid dependency. Always refer to a named example for each strategy.